PLATE NO. VIII · TRANSLATION GEOMETRY

SPECIMEN: DPT-2028 · ARTICULATED KINEMATICS

DATUM: 40°01'14"N 75°09'22"W [GLENSIDE TERRESTRIAL]

INSTRUMENTATION: 7-DOF SERIAL MANIFOLD × RIEMANNIAN PHASE SPACE

METRIC TENSOR: $g_{ij} = J_i^T(q)\, M(q)\, J_j(q)$ · INVARIANT TORSION

KINEMATIC MONISM

THE SPATIAL GEOMETRY OF ARTICULATED MANIFOLDS

A HARMONIC SYNTHESIS OF HUMAN BIOMECHANICS & INDUSTRIAL ROBOTIC DEGREES OF FREEDOM

SYSTEM METRIC: SO(3) × SE(3)

DEGREE OF FREEDOM: 07 [$q_1 .. q_7$]

ANGULAR ENVELOPE: 0.00° → 135.24°

VELOCITY VECTOR: $\dot{q}(t) \in C^2[0, T]$

STATUS: VERIFIED MATHEMATICAL STUDY

FIG. 01 · SAGITTAL PROJECTION

TIBIOFEMORAL RANGE: 0°–135°



ICR OFFSET: +3.2mm

FIG. 02 · ORTHOGONAL SO(3)

EULER ROTATION MATRIX [α β γ]



DET(R) = +1.00000

DENAVIT-HARTENBERG COORDINATE CHAIN

KINEMATIC LINK TRANSFORMATION MATRICES

LINK 01	θ_1 = -045.0°	θ: 0° | d: 0mm
LINK 02	θ_2 = -016.5°	θ: 90° | d: 45mm
LINK 03	θ_3 = +012.0°	θ: 0° | d: 90mm
LINK 04	θ_4 = +040.5°	θ: 90° | d: 135mm
LINK 05	θ_5 = +069.0°	θ: 0° | d: 180mm
LINK 06	θ_6 = +097.5°	θ: 90° | d: 225mm
LINK 07	θ_7 = +126.0°	θ: 0° | d: 270mm



FIG. 03 · PHASE SPACE ATTRACTOR

ANGULAR DISPLACEMENT vs. VELOCITY [θ vs. $\dot{\theta}$]

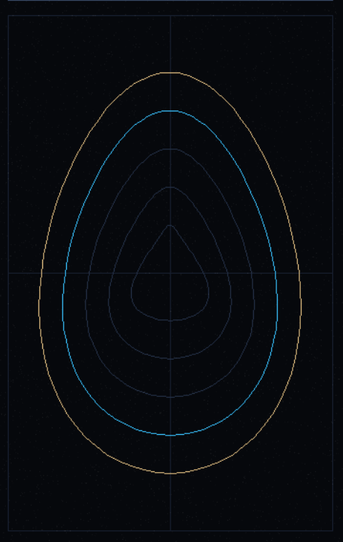


FIG. 04 · GROUND REACTION FORCE HARMONICS

FOURIER SPECTRAL DECOMPOSITION OF STRIDE CADENCE

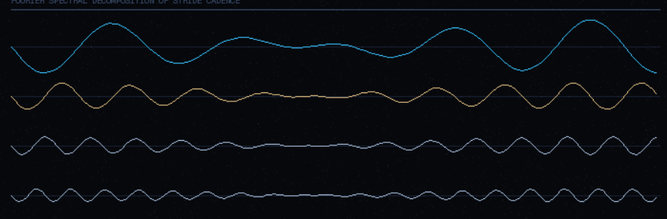


FIG. 05 · SERIAL TORQUE TRANSMISSION

ACTUATOR POWER vs. MUSCULOSKELETAL MOMENT ARM



[AXIOM 01 · RIEMANNIAN KINEMATICS]

Space and form do not merely contain movement;
they are the crystallized residual trace of motion itself.
The boundary between mechanical serial chains and biological limbs
dissolves into continuous vector manifolds and invariant coordinate strata.

$$d/dt(\partial L/\partial \dot{q}) - \partial L/\partial q = \tau - J_c^T(q)\cdot\lambda \quad | \quad \nabla H(p,\ q) = 0$$

ARCADIA MOVEMENT SCIENCE

[AXIOM 02 · KINETIC EQUILIBRIUM]

Every articulation obeys an overarching harmonic law.
In biological structures and artificial manipulators alike,
force and compliance achieve resolution without decorative excess.
Information lives in design, precision, and spatial cadence.

$$T_7^0(q) = \prod A_i(q_i) \quad | \quad \mathrm{rank}(J(q)) = 6 \quad | \quad \det(M) > 0$$

PLATE NO. IX · BIOMECHANICAL ARTICULATION

SPECIMEN: ANATOMICAL FEMUR-TIBIA JOINT (CC BY 4.0 DBCLS)

INSTANTANEOUS HELICAL AXIS: J-CURVE EVOLUTE MANIFOLD

FEMORAL ROLLBACK: +18.4mm AP TRANSLATION OVER 135° FLEXION

CONTACT STRESS TENSOR: $\sigma_{ij} = C_{ijkl} \cdot \varepsilon_{kl}$ (MEDIAL/LATERAL)

TIBIOFEMORAL KINEMATICS

INSTANTANEOUS HELICAL AXES & CONTACT STRESS MANIFOLDS

A PARAMETRIC STUDY OF HUMAN KNEE ROLLBACK, LIGAMENT STRAIN & SCREW-HOME ROTATION

SCREW-HOME: 15° EXT ROT

CRUCIATE BUNDLES: ACL / PCL

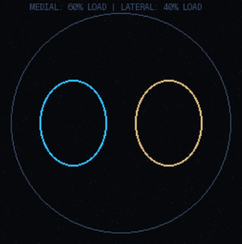
MENISCAL COMPLIANCE: E = 12 MPa

CONVERGENCE: FOUR-BAR LINKAGE

ARCADIA DPT KINELAB SPECIMEN

FIG. 01 · PLATEAU CONTACT ISOBARS

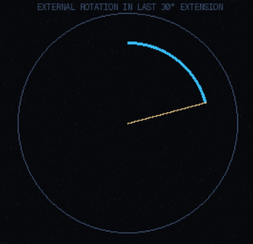
MEDIAL: 60% LOAD | LATERAL: 40% LOAD



PEAK STRESS: 3.4 MPa

FIG. 02 · SCREW-HOME TORSION

EXTERNAL ROTATION IN LAST 30° EXTENSION



ANGULAR ROTATION: +15°

ELECTROMYOGRAPHIC ENVELOPES

NORMALIZED MUSCLE ACTIVATION CYCLES (% MVC)

RECTUS FEMORIS	82% MVC
VASTUS LATERALIS	94% MVC
VASTUS MEDIALIS	91% MVC
BICEPS FEMORIS	68% MVC
SEMITENDINOSUS	74% MVC
GASTROCNEMIUS MED	62% MVC
TIBIALIS ANTERIOR	55% MVC

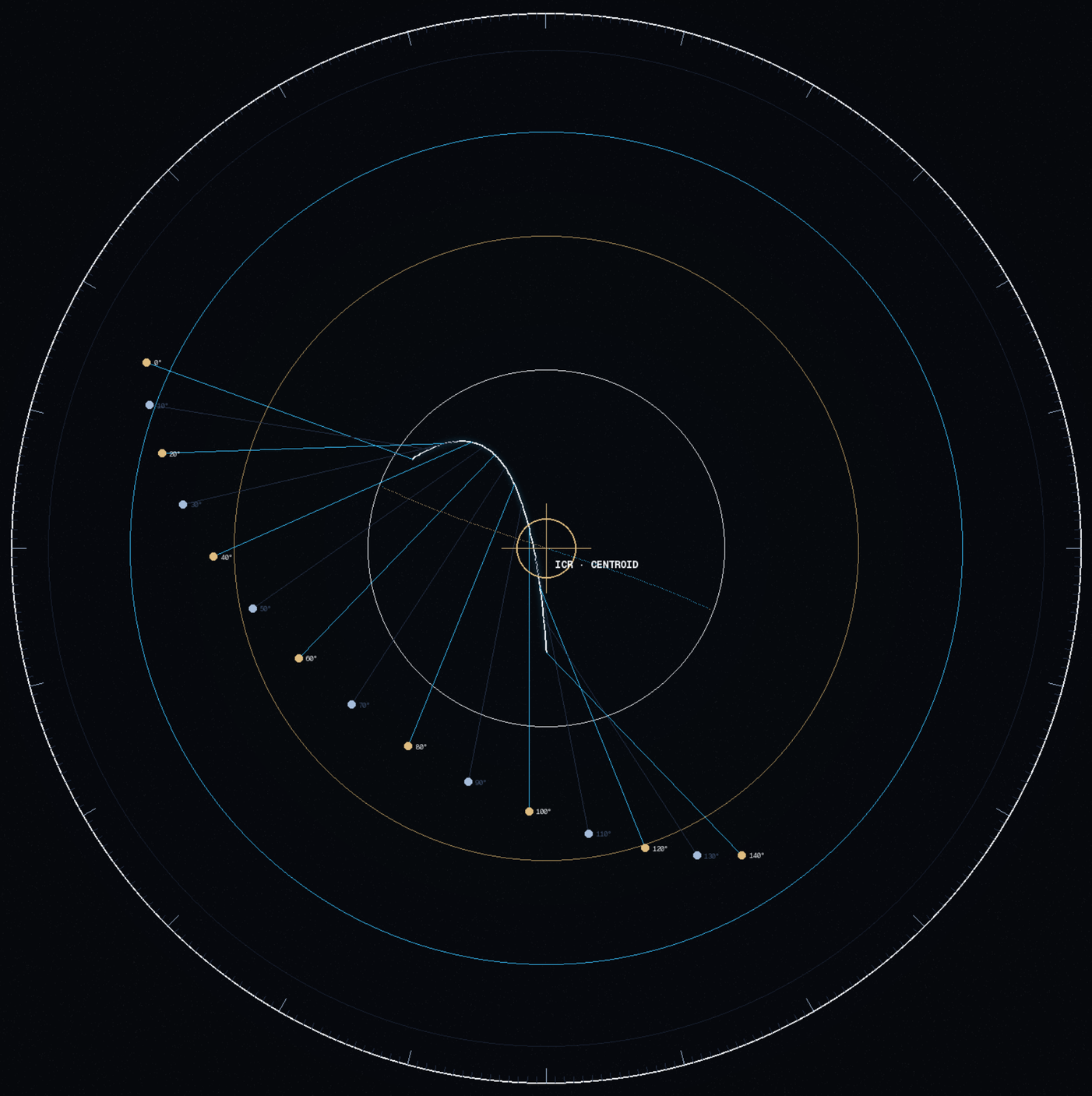


FIG. 03 · PATELLOFEMORAL FORCE

JOINT REACTION FORCE vs. SQUAT DEPTH

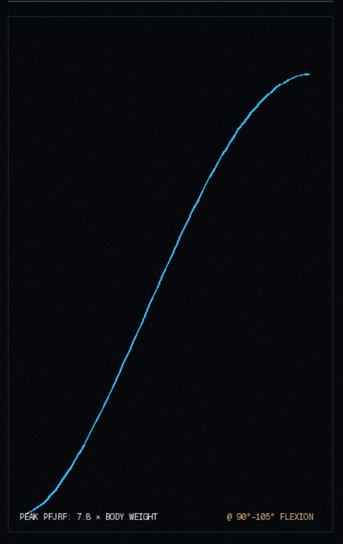


FIG. 04 · MENISCAL HOOP STRESS HOOKES LAW

CIRCUMFERENTIAL TENSILE FIBER RECRUITMENT

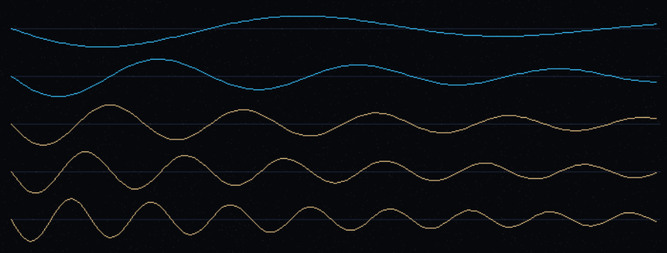


FIG. 05 · VISCOELASTIC HYSTERESIS LOOP

ENERGY DISSIPATION IN ARTICULAR CARTILAGE



[AXIOM 03 · BIOLOGICAL OPTIMIZATION]

The knee joint is an asymmetric four-bar linkage in continuous equilibrium.
Articular surfaces conform under load to minimize shear energy dissipation.
Femoral condylar geometry balances mobility with terminal screw-home stability.
Bone mineral density models dynamically in response to applied stress tensors.

$\sigma = E \cdot \varepsilon + \eta \cdot d\varepsilon/dt$ | WOLFF'S LAW CONTINUUM FORMULATION

ARCADIA BIOMECHANICS LAB

[AXIOM 04 · ANATOMICAL CONTINUUM]

Every biological movement traces an invariant mathematical path.
From muscular motor units to ground reaction vectors,
human gait embodies evolutionary architectural efficiency.
Kinematic intelligence resides within morphological structure itself.

$ICR(t) = v(t) \times \omega(t) / ||\omega(t)||^2$ | INVARIANT ROTATION AXIS

PLATE NO. X · WORKSPACE TOPOLOGY

SPECIMEN: NVIDIA SIMREADY FRANKA RESEARCH 3 (CC BY 4.0)

MANIPULABILITY INDEX: w = √(det(J(q)·J^T(q))) = 0.142

CARTESIAN REACHABLE WORKSPACE ENVELOPE: RADIUS = 855mm

SINGULARITY HULL: det(J) = 0 [WRIST/SHOULDER ALIGNMENT]

CYBERNETIC MANIPULATION

7-DOF JACOBIAN MANIPULABILITY & SINGULARITY HULLS

A TOPOLOGICAL INVESTIGATION OF ISOTROPIC DEXTERITY, TORQUE LIMITS & END-EFFECTOR VELOCITIES

PAYLOAD: 3.0 kg RATED

REPEATABILITY: ±0.1mm

CONTROL CYCLE: 1000 Hz

TORQUE SENSORS: 7 × STRAIN

NVIDIA SIMREADY CENTRAL

FIG. 01 · JOINT VELOCITY SATURATION

LIMIT: ±150°/s [JOINTS 1..4] | ±180°/s [5..7]



MAX LIN SPEED: 1.7 m/s

FIG. 02 · IMPEDANCE STIFFNESS

CARTESIAN STIFFNESS TENSOR: K_x, K_y, K_z



K_t: 3000 N/m | K_r: 300 Nm/rad

FORWARD KINEMATICS CHAIN

SE(3) HOMOGENEOUS MATRIX MULTIPLICATION

T_1^0 — R_1 ∈ SO(3) | p_1 ∈ ℝ³
DH: a=0mm, α=0°, d=120mm

T_2^1 — R_2 ∈ SO(3) | p_2 ∈ ℝ³
DH: a=0mm, α=90°, d=140mm

T_3^2 — R_3 ∈ SO(3) | p_3 ∈ ℝ³
DH: a=0mm, α=0°, d=160mm

T_4^3 — R_4 ∈ SO(3) | p_4 ∈ ℝ³
DH: a=0mm, α=90°, d=180mm

T_5^4 — R_5 ∈ SO(3) | p_5 ∈ ℝ³
DH: a=0mm, α=0°, d=200mm

T_6^5 — R_6 ∈ SO(3) | p_6 ∈ ℝ³
DH: a=0mm, α=90°, d=220mm

T_7^6 — R_7 ∈ SO(3) | p_7 ∈ ℝ³
DH: a=0mm, α=0°, d=240mm

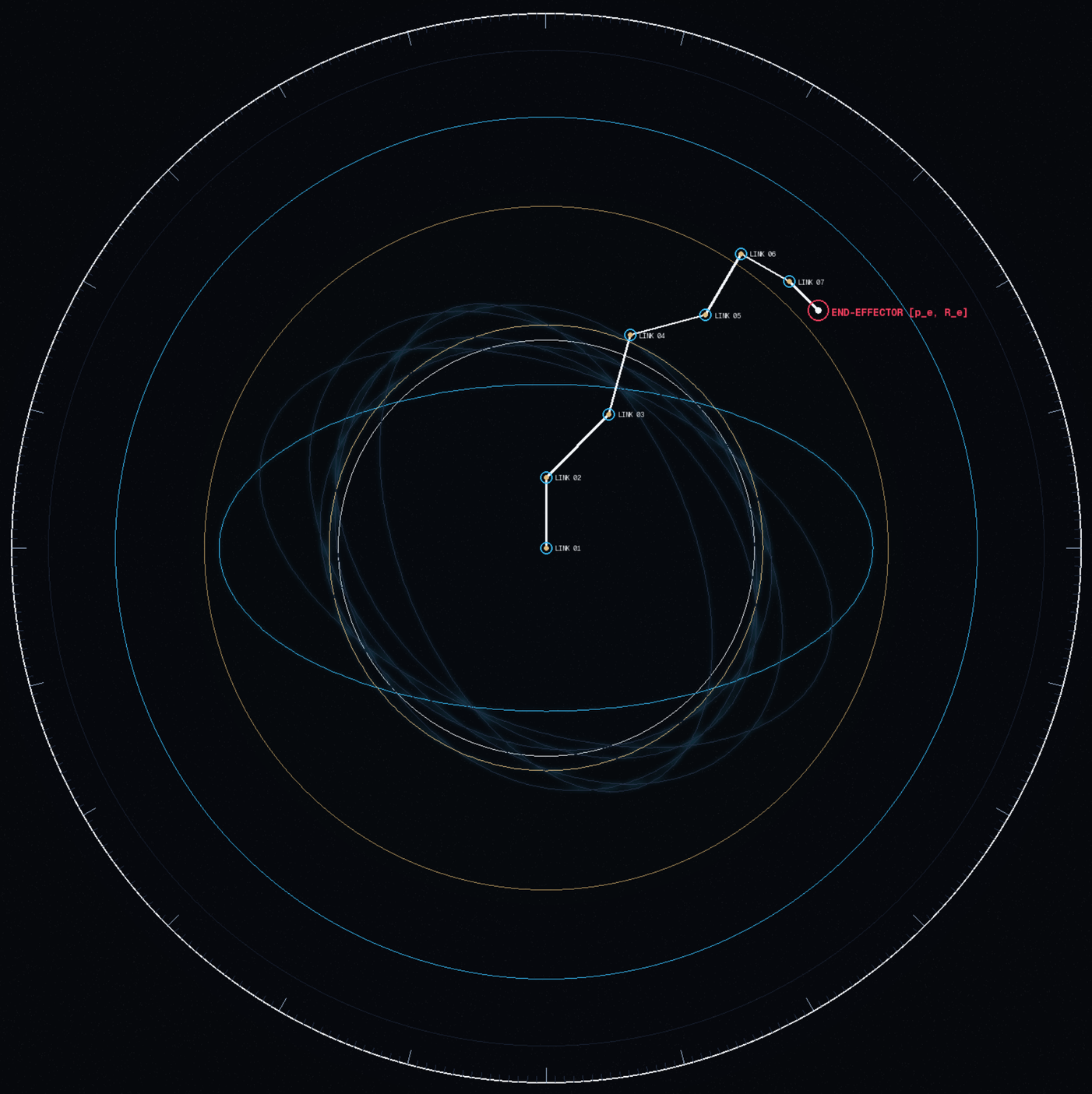
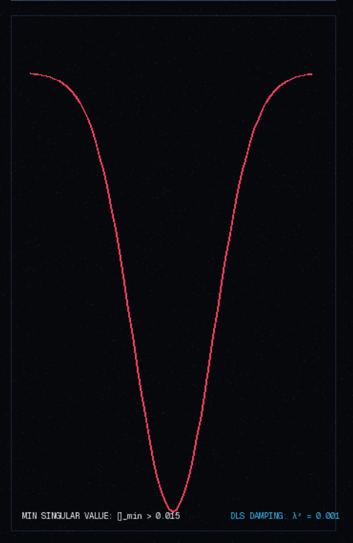


FIG. 03 · SINGULARITY HULL

DAMPED LEAST SQUARES REGULARIZATION



MIN SINGULAR VALUE: σ_min > 0.015 — DLS DAMPING: λ² = 0.001

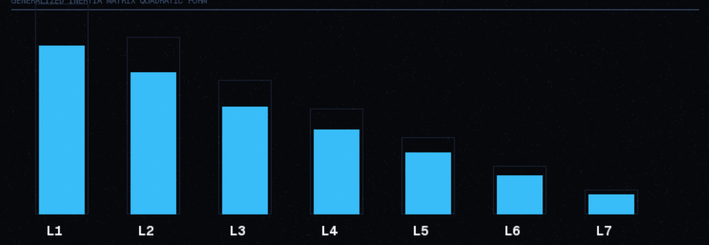
FIG. 04 · ACTUATOR BODE RESPONSE

FREQUENCY RESPONSE OF CLOSED-LOOP CURRENT CONTROLLER



FIG. 05 · KINETIC ENERGY DISTRIBUTION

GENERALIZED INERTIA MATRIX QUADRATIC FORM



[AXIOM 05 · DIFFERENTIAL ROBOTICS]

The manipulator mapping between joint rates and Cartesian space is governed by the linear Jacobian operator v_e = J(q) q̇.
Through redundant degrees of freedom, the null-space projection optimizes joint torque and posture without altering end-effector velocity.

q̇ = J^#(q)·v_e + (I - J^# J)·q̇_0 | NULL-SPACE INVARIANCE

NVIDIA SIMREADY & ARCADIA

[AXIOM 06 · CONSERVATION OF MOMENTUM]

Generalized momentum in articulated serial chains conforms to closed Hamiltonian dynamics across all operational manifolds.
Torque saturation boundaries demarcate physical interaction safety.
Real-time cybernetics bridges digital simulation with physical reality.

M(q)·q̈ + C(q, q̇)·q̇ + g(q) = τ | EULER-LAGRANGE ROBOTICS